

THEORY OF COMPUTATION (CS302)

Halting Problem & Decidability

Learning Objectives

Understand decidable and undecidable problems, recursive and recursively enumerable languages, the Halting Problem, and the Church-Turing Thesis.

Introduction

Decidability is a fundamental concept in computation theory that determines whether a problem can be solved by an algorithm in finite time.

Decidable Problems

A problem is decidable if there exists a Turing Machine that always halts and correctly determines whether a given input belongs to a language.

Examples of Decidable Problems

- DFA Acceptance Problem
- DFA Emptiness Problem
- Finite Language Membership Problems

Undecidable Problems

A problem is undecidable if no algorithm can be constructed that always produces a correct yes/no answer for every possible input.

Recursive Languages

A language is recursive if there exists a Turing Machine that always halts and decides membership for every input string.

Recursively Enumerable Languages

A language is recursively enumerable if a Turing Machine accepts strings belonging to the language, but may not halt for strings not belonging to the language.

Halting Problem

The Halting Problem asks whether a program will halt or continue running forever for a given input. Alan Turing proved that no general algorithm can solve this problem for all possible programs and inputs.

Proof Idea of Halting Problem

- Assume a machine H exists that solves the Halting Problem.
- Construct another machine using H .
- Apply self-reference.
- A contradiction occurs.
- Therefore H cannot exist.

Church-Turing Thesis

The Church-Turing Thesis states that any computation that can be performed algorithmically can be computed by a Turing Machine.

Applications

- Compiler Design
- Artificial Intelligence
- Program Verification
- Complexity Theory
- Formal Language Theory

Comparison: Recursive vs Recursively Enumerable

Recursive Languages: Always halt and decide.

Recursively Enumerable Languages: May not halt for some inputs.

Exam Important Questions

1. Define Decidable Problem.
2. Define Undecidable Problem.
3. Explain Recursive Languages.
4. Explain Recursively Enumerable Languages.
5. State and explain the Halting Problem.
6. Explain the Church-Turing Thesis.

Summary

The Halting Problem demonstrates the limitations of computation. Decidability theory helps classify problems based on whether they can be solved algorithmically, forming a core part of theoretical computer science.